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Investigation into the Identification of Ragas: A Detailed Analysis

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***Abstract*—This paper endeavors to examine and ascertain the optimal AIML methodologies for the discernment of Ragas in Indian Classical Music. Following an extensive review of pertinent literature, we have meticulously devised a methodology for the comprehensive processing of our data sets, aiming to yield optimal results. The overarching goal of our paper is to delve into diverse AI/ML techniques, initially applied to the data sets under consideration with constrained values, and subsequently broaden the data set scope, to encompass all 5040 Ragas in Indian Classical Music, thereby enhancing system efficiency. Our investigation offers valuable insights into potential enhancements to the data set, addressing its limitations and augmenting the concept's efficacy for prospective applications. The data set necessitates meticulous pre processing, paving the way for the strategic implementation of techniques such as K-nearest neighbors (KNN) and Convolutional Neural Networks (CNN) on the Hindustani Music Dataset(HMD) and Classical Music data set (CMD)separately, with the ultimate objective of extracting optimal outcomes.**

***Index Terms*—AI/ML, Raga, Classification, Deep Learning**

I. Introduction

Indian classical music, encompassing Marg Sangeet and Shastriya Sangeet, consists of two major traditions: Hindustani (North) and Carnatic (South). Instruments vary, with Sitar and Tabla in Hindustani, and Veena and Mridangam in Carnatic. Both traditions showcase rich histories rooted in ancient Indian musical traditions, with global recognition for their profound impact on the classical music landscape. Indian classical music centers on Raga (melodic structure) and Tala (rhythmic cycle). Ragas, woven from Swara (notes, including microtones), provide a diverse melodic palette, guiding artists in creating intricate melodies [1]. Tala sets the rhythmic framework for improvisation. This distinctive approach distinguishes Indian classical music from Western classical traditions [2].

A. Raga

A Raga, a cornerstone of Indian music, encapsulates note intonation, duration, and order, akin to words forming phrases, crafting expressive atmospheres. With a set of notes on a scale, Ragas influence mood and aesthetic sentiment. The classical tradition, though theoretically allowing thousands, refines choices to several hundred key Ragas [3]. Closely intertwined with Tala, guiding time division, Ragas play a pivotal role in classical Indian music. A Raga, the structured Swara sequence representing mood and sentiments in Indian Classical music genre (Carnatic and Hindustani), is a fundamental element [4]. Indian music basically has seven Swaras (notes): Sa, Ri, Ga, Ma, Pa, Dha, Ni. There are hundreds of Raga varieties in Indian Carnatic music. These are derived from 72 parent or Janaka Ragas, formed by combining 12 Swarasthanas. The sheer number, intricate structure, and subtle variations make Ragas difficult to identify even for humans without years of practice [5].

B. Tala

In Indian music, a Tala encompasses the entire domain of musical meter, providing a structured beat framework for composition and performance. The measuring medium of the speed of music compositions is Tala. Talas comprises angaas called as Taalangaas. These angaas are the bodies or parts of Tala. These angaas are shown in the counting of Talas by hand gestures.

II. Related Work

Research, as a systematic and scientific process, relies on research methodology to guide its systematic, scientific, and objective execution. Different approaches have been researched on for identifying Ragas in music. One method involves directly transcribing Ragas into Swaras at regular intervals and using classifiers like K-NN or SVM [6]. Another approach likens Raga classification to word recognition, treating Raga compositions as words formed from Indian classical music notes. This method utilizes a Hidden Markov

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Model (HMM) [7], to learn note transitions based on well-defined sequences but heavily relies on accurate initial note transcription [8]. However, it overlooks the essential pitch distribution information crucial for effective Raga identification. In this research, a unique approach employing deep learning and signal processing is introduced to identify Ragas from audio, specifically utilizing raw spectrograms. The proposed method, which includes pre processing steps and models, demonstrates an impressive testing accuracy of 98.98%, on a subset of 10 Ragas within the CompMusic data set. The study comprehensively explores various hyper parameters, sound source separation, and silent part removal, presenting detailed results and findings. The discussion delves into predictions and the performance of deep learning models on audio beyond the data set. This innovative technique exhibits promise, enabling real-time predictions from raw audio or recordings, with potential applications in Music Information Retrieval, content-based filtering, and teaching-learning domains. In the study authored by Hiteshwari Sharma and Rasmeet S. Bali [9], various machine learning classifiers are utilized to distinguish and categorize different Ragas. This is based on key parameters such as choice of notes, Arohana-Avarohana, Pakad, and Gamakas. The K-star algorithm is used for demonstrating superior performance and has accuracy of 93.38%, surpassing other classifiers. The classifiers were evaluated on Ragas Des, Bhupali, Yaman, and Todi each belonging to distinct Thaats or musical scales. Feature extraction involved the use of Chromogram, with Vadi Swara playing a crucial role in Raga identification. The results showcased the effectiveness of K-star is used in achieving high accuracy, supported by a Kappa statistic of 0.912, indicating near-perfect classification. This detailed study contributes important insights into the application of machine learning for Raga recognition, emphasizing K-star's notable performance in capturing the intricate patterns of Indian classical music [9].

A study by Ankit Anand [10],introduces a pioneering approach to Carnatic music analysis, departing from conventional feature engineering methods. Instead, a Convolutional Neural Network (CNN) is employed to discern distinctive Raga characteristics directly from a song's predominant pitch values, a departure from CNN's usual application in visual imagery analysis. The model exhibits promising performance on standard data sets, achieving accuracies of 96.7% and 85.6% for five and eleven Ragas, respectively. Notably, the CNN's adaptability is tested on allied Ragas, suggesting potential broader applicability. This innovative method signifies a shift towards automated Raga recognition in Carnatic music through deep learning techniques.

Ekta Patel and Savita Chauhan present a comprehensive analysis of AI/ML methods applied to Raga detection in music, highlighting previous techniques and their limitations [9]. It emphasizes the importance of key phrase identification, consideration of social behaviors, and the incorporation of pitch and mood identification in training subsets. The proposed work involves extracting music features using the MIR Toolbox in MATLAB, arranging them in ARFF file format, and utilizing the WEKA tool for Raga detection. The study highlights the significance and importance of discretization in improving classifier accuracies.

Dipti Joshi, Dr. Jyoti Pareek, Pushkar Ambatkar, address the challenges of preprocessing audio files in Hindustani Classical music, considering factors such as multiple instruments, relative scales of notes, and the dynamic nature of Raga compositions[8]. Feature extraction involves the use of Power Spectrogram, Mel-Frequency Cepstral Coefficients (MFCC), Spectral Centroid, Zero-Crossing Rate, and Spectral Bandwidth. The dataset comprises 60-second audio clips, which are then divided into frames for targeted analysis of specific song elements. The study employs the Librosa and Scikit Learn libraries for feature extraction and algorithm implementation. Results indicate that KNN, with varying neighbor values and an 80/20 Train/Test ratio, achieves higher accuracy compared to SVM, highlighting the effectiveness of KNN in Raga classification [7].

III. Prominent Research Methodologies

This section of the paper describes in detail the 'Pitch Co-occurrence' method that makes use of Co-occurrence matrices and Confusion matrices for Raga identification. As insightfully explained by Dr. Mrs. Kalyani R. Joshi and Mrs. Vibhavari Rajadnya in their paper titled Raga Classification Based on Novel Method of Pitch Co-Occurrence , this research has utilized the pattern developed due to co-occurrence of pitches of Swaras in audio files obtained recordings in the CMD dataset (Publicly available database consisting of 480 recordings of 40 Ragas) for classification [11]. This pattern gives a concise representation of the signal which contains time and frequency information of the Raga. K Nearest Neighbor (KNN) has been used as the classifier to determine the class label (Raga name) of a given Raga. This paper majorly emphasizes on second order statistical measures to determine feature vectors each audio file so that the can be used as input for KNN Classification. As demonstrated by the experimental result, which yields an accuracy of 78.1%, the suggested approach is moderately effective [9].

Paper authored by Dr. Mrs. Kalyani R. Joshi and Mrs. Vibhavari Rajadnya ,uses gray scale plots of Co-occurrence matrices determined for each audio file in the HMD dataset (Publicly available database consisting of 300 recordings of 30 Hindustani Ragas consisting of 130 hours of audio recordings stored as 160 kbps mp3 files which is part of Comp Music project as the basis for classification using K Nearest Neighbor (KNN) method [12].

This paper emphasizes on the fact that each Raga will have a unique spectral power distribution and uses this as an important criterion to develop co-occurrence matrix plots for each Raga in order to use them for classification. Spectral Power Distribution (SPD) refers to the power per unit area per unit wavelength of any radiometric quantity. The experimental results show that that this system is effective, yielding a 93.7% accuracy rate. Cursory literature review of various other works in this domain by researchers' show that most of them fail

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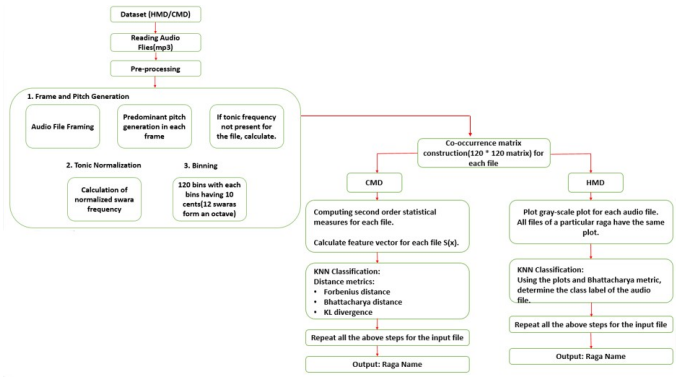

Fig. 1. Raga Classification Based on Pitch Co-Occurrence Method

to capture both the tonal and temporal aspects of a Raga. Temporal aspects can majorly refer to instances where two Ragas can have the same set of notes and but different in the gamakas or the time spent on each note. The above-mentioned classification methodologies address these issues using Co-occurrence matrix concepts.

The following subsections of this section gives a detailed explanation of the above discussed approaches for Raga classification.Fig.1 represents the overall process . Step 1 that majorly deals with pre processing of audio files and construction of co-occurrence matrices for the audio files. This is common for both the approaches. While the methodology using the CMD dataset uses second order statistical measures for classification (described in step 02), the second paper [12] makes use of the results of step 1 to develop plots for Co-occurrence matrices used for classification. The KNN classification methodology described in step 3 is again, common for both approaches except for the fact that the parameters being used for the classification are different.

A. Methodology

1. Tonic Normalization and constructing Co-occurrence matrix:

- Each audio file is divided into a number of frames. Each of these frames will have a pitch value corresponding to it.
- Each audio file will be having a tonic frequency contained in a file.
- The tonic frequency or the Swara Sa is the base for any composition. All other Swaras for a composition are tuned relative to this tonic frequency.
- Using the tonic pitch provided in the database being used, the tonic normalization is carried out to get the normalized pitch values for an audio file in Hertz.
- These normalized pitch values are converted to cents scale from frequency scale to musically relevant as 1200 cents correspond to one octave [13]. The normalized Swara Frequency is given by (2).

$$c_i = 1200 log_2(f_i/w) \quad (1)$$

- Here, c_i Normalized Swara frequency in cents, f_i Predominant pitch in the particular frame, w is Tonic pitch of the lead artist in the audio file
- All the 1200 cents are then mapped into 120 bins with each bin containing 10 cents. This can be considered as optimal pitch resolution. This is called as binning where the data is divided set of intervals and each bin is replaced with a single computed value for that particular bin using the values in the bin. Each bin value gets replaced with the calculated single value.

$$B(x) = [(x\eta/1200)] mod \eta \quad (2)$$

Where B the octave-wrapping integer for that particular bin, is given by (2).

- Using these 120 bins generated for each audio file, a 120*120 Co-occurrence matrix is constructed.
- In any co-occurrence matrix, the element will represent the occurrence of a pair under consideration in a particular defined context.
- In the co-occurrence matrix considered here the pairs are the Swara/bin pairs and the context under consideration are two successive frames which are some distance apart. So, each element of this particular co-occurrence matrix will represent how frequently a bin pair will occur in two successive frames of an audio file [14].

2. Evaluation of Second Order Statistical measures:

- The next major step is to calculate the second order statistical measures of the co-occurrence matrix for each audio file. This includes: homogeneity, energy, correlation, contrast and entropy.
- Homogeneity: Homogeneity is a measure which indicates how close the elements in the matrix are with reference to its diagonal.
- Contrast: It computes the difference in pitch values between two neighboring frames in the entire recording file. If it is constant value, contrast will be 0.
- Entropy: It is an indication of variation in pitch values in frames. If the pitch is constant, all thru the files, it will be 0.
- It calculates addition of squared elements. It is an indication of how similar are the pitch value levels in adjacent frames. If the pitch is constant, it will return 1.
- Correlation: Measure of how correlated a bin value (which represents normalized pitch in this project) in one frame is to its neighboring frame in an mp3 file. If the frames are positively correlated it will be 1 else -1. These parameters were calculated for various co-occurrence matrices calculated by introducing delays between frames. The delays introduced were in the range 0.1s to 1.5s with 0.1 seconds as step. For each of the step in the given range, a co-occurrence matrix and the five parameters were calculated.

3. Classification using KNN:

- Using the parameters and the co-occurrence calculated at each step, feature vector is constructed. This feature vector is used for the Raga classification using KNN [15].

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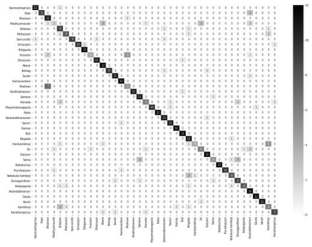

Fig. 2. Confusion Matrix on CMD

- This feature vector is calculated for each audio file in the database.
- KNN algorithm is used for classification.

The distance between two audio recordings with feature vectors S(m) and S(n) are calculated using three prominent distance metrics as seen in (3),(4),(5) and (6).

- Frobenius Norm

$$D_F^{(N,M)} = ||S_n - S_m||_2 \tag{3}$$

Where S_n is the nth audio recording's feature vector, S_m is the mth audio recording's feature vector

- Kullback Leibler Divergence

$$D_{KL}^{(N,M)} = D_{KL}(S^{(n)}, S^{(m)}) + D_{KL}(S^{(m)}, S^{(m)}) \tag{4}$$

$$D_{KL}^{(X,Y)} = \sum X\log(X/Y) \tag{5}$$

- Bhattacharya distance

$$D_B^{(n,m)} = -\log(X\sqrt{S^{(n)}} \cdot \sqrt{S^{(m)}}) \tag{6}$$

Here, each element of the feature vector of a particular audio file is considered for operation and then all these elements on which the operation has been performed are added. The K values considered were 1, 3 and 5.

IV. Results

For the paper making use of the CMD dataset, the results suggest that the difficulty of the Raga categorization problem increases with the number of classes [16]. It is evident that the pattern for contrast and entropy features contribute most to the classification of Ragas. The large variety in pitch values at different delays can be explained by entropy. Pitch values differ by contrast, which is observed to be greater for particular delays. This is a result of the Raga's avarohana and arohana using distinct sets of svaras. The melodic shifts between svaras are likewise well captured by these co-occurrence matrix SOS feature patterns. The correlation between the classifier's actual and predicted classifications is shown in the confusion matrix depicted in Fig. 2. It is calculated over the anticipated classes in order to ensure accuracy. This methodology yields an accuracy of 78.1%. The leave-one-out validation strategy has been applied. In this, the remaining recordings are used for training, and one recording is used for testing. This is used by both the approaches using CMD and HMD data sets [17].

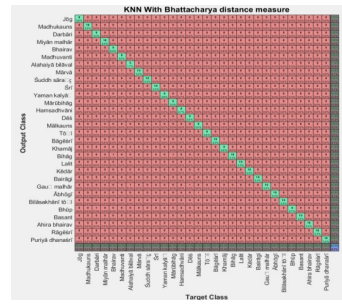

Fig. 3. Confusion Matrix on HMD

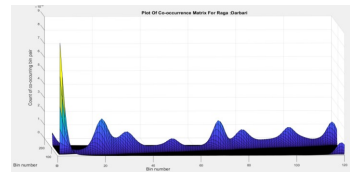

Fig. 4. Plot of Co-Occurrence matrix for Raga Darbari (HMD)

From the second paper on HMD, it can be inferred that every Raga has a different distribution of spectral power, or pitch frequencies, and it has been found that a co-occurrence matrix, which represents the distribution pattern, is a useful tool for classifying Ragas. The distribution is more diagonally oriented. These line up with the svaras. The prominent components on the diagonal line up with the Vadi svara, which is the one that is used or stressed out the most. Different sets of svaras are used in the Raga's arohana and avarohana, which results in an uneven spread with respect to the diagonal. The co-occurrence matrix has been found to be an effective tool for capturing the short-term temporal relationship between pitches[14]. For accuracy purposes, a confusion matrix is computed over the predicted classes which is depicted in Fig. 3. Based on the majority class of its k-nearest neighbors (K=1, 3, 5), the predicted class is assigned. In the event of a tie, the majority class is chosen at random.

Fig. 4 and Fig.5 illustrates the Unique plots developed for each Raga in HMD. This methodology yields an accuracy of 93.7. Similarly, in a study by Ankit Anand [10], a Convolutional Neural Network (CNN) model has been trained to extract specific features from an audio file and subsequently classify it into a specific Raga class label [18].

The process can be further divided into 2 parts:

- Data pre-processing: As seen in the Fig. 6,in data pre-processing, the tonic pitch of the audio file is extracted

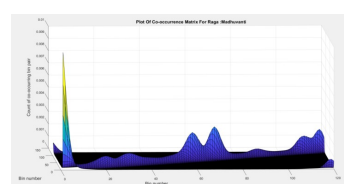

Fig. 5. Plot of Co-Occurrence matrix for Raga Madhuvanti (HMD)

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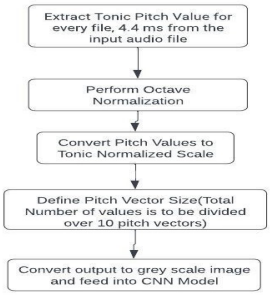

Fig. 6. Data Pre-Processing for CNN Input Generation

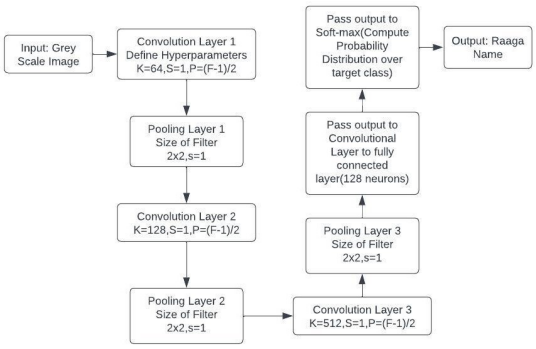

Fig. 7. Raga Classification using Convolutional Layers and Softmax

for every 4.44 ms. This data is then 'octave normalized' to bring all input values to a common pitch. Pitch vectors are then defined such that the audio file is divided evenly over 10 pitch vectors. This data is then converted into a gray-scale image and fed into the CNN model as input[16].

- Classification: The CNN model used has 3D volume with neurons in each layer connected to a small portion of neurons from the previous layer illustrated in Fig. 8.

As seen in the Fig. 7, the components of this network and their working is as described below:

- Input Layer: Input to the model is the gray scale image of the tonic normalized pitch values of the input audio file. (CNN usually works on image data and hence comes the necessity to represent the audio file as a gray scale image) [19].
- Convolutional Layer: There exist 3 convolutional layers which has a set of learnable filters, with a small spatial window is extended to filter action, extending through the entire depth when we consider the input volumes. 4 hyper parameters or external configuration variables used to manage the model training are: 1.K – depth of filter 2.S – filter stride 3.P- number of zero padding 4. F- filter size Given an input volume of size

$$W_i \times H_i \times D_i$$

and a set of convolutional filters of size FxF, the output volume is determined by the relation:

$$W_0 = (((W_i - F + 2 * P)/S) + 1 \quad (7)$$

$$H_0 = ((H_i - F + 2 * P)/S) + 1 \quad (8)$$

$$D_0 = K \quad (9)$$

Where, the output size of the Convolutional Layer, W_o and H_o is given by (7) and (8). The depth or number of channels in the output feature map D_0, is given by (9).

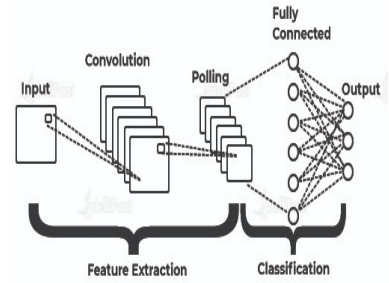

Fig. 8. Convolutional Neural Network Data flow

Table 1: Performance of approach on D1 and D2 on F value

F	Accuracy of D1 (in%)	Accuracy of D2 (in%)
2	95.8	79.2
3	95	83.1
4	96.7	85.6

The 3 convolutional layers have their depths progressively increasing in the order of 64, 128 and 512.

- Pooling: The total number of parameters and computations have to be kept in check, then we make use of max-pooling at each level , the maximum value is returned in a window in an input layer. Pooling filters are of the size 2*2 and stride 1 are made use of at every layer.
- Fully Connected Layer and Softmax: Convolutional layers output is fed into a completely connected layer consisting of 128 neurons in total. The soft max layer which is the fully connected last layer has one node for each target class. The probability distribution is computed by soft max, for given input over all target classes and the input is classified accordingly [18].

V. Experimental setup

A. Dataset Overview:

- The dataset comprises 480 full-length recordings, encompassing 40 Ragas.
- Each Raga includes 12 songs, and there are 6 features recorded per song.

B. Evaluation Subsets:

- D1: Consists of 5 non-allied Ragas—Ritigaula, Nata, Sankarabharanam, Todi, and Varali
- D2: Encompasses 11 distinct Ragas—Ritigaula, Nata, Sankarabharanam, Todi, Varali,
- AD1: Comprises 2 allied Ragas—Madhyamvati and Sri.
- AD2: The 2nd set of allied Ragas, including Bhairavi and Mukhari.

C. Data Generation:

- Each Raga across all datasets contributes 12 songs, resulting in the generation of 120 images per Raga.
- Gray scale images sized 512 x 128, depicting the variation in pitch over time, serving as the input to the Convolutional Neural Network (CNN).

D. Data Splitting:

- Across all datasets (D1, D2, AD1, AD2), the data is partitioned into training, testing and validation sets which is used to maintain a ratio of 70:15:15.

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Table 2: Performance of approach on allied Ragas for different values of F(accuracy%)

F	AD1 (in%) MadhyamavatiSri	AD2(in%) Bhairavi-Mukhari
2	95.8	79.2
3	95	83.1
4	96.7	85.6

The model's accuracy in classifying audio inputs from D1 and D2 datasets is tabulated in Table 1 and Table 2. Deep artificial neural networks face challenges, such as the demand for extensive data and significant computational power to handle numerous parameters. CNNs, specifically, experience reduced accuracy as class numbers grows, accompanied by the risk of over fitting. These issues can be addressed by leveraging standardized datasets, offering enhanced results with a more extensive repertoire of Ragas. Overcoming limitations in data quantity and computational demands, a standardized dataset proves crucial for optimizing the performance of deep learning models, particularly in scenarios with a larger number of classes like diverse Ragas.

VI. Conclusion

The automated identification of the Raga within an Classical song under the Indian classical music genre holds significant applications, particularly used in areas such as recommendations of music and indexing. This process facilitates the categorization and recommendation of music based on specific Ragas, contributing to a more refined and personalized musical experience for listeners. The efficiency of the classification is based on the processing of the audio files. The more precise the data set with a greater number of training data the better will be the results received on feeding the testing data It also finds applications in musicological studies, similarity-based search. Hence, it is very essential to develop relevant techniques to identify the Ragas in an audio file in the most convenient way. With this first of its kind application, we intend and wish to provide a reliable application or methodology that can fill the gap currently existing between the needs of musicians and other associated artists and what the current applications or technology has to offer.

VII. Future Scope:

Artificial Intelligence (AI) and Machine Learning (ML) have the potential to revolutionize Raga detection by automating the traditionally labor-intensive process. In conventional methods, human experts manually identify and label musical features, introducing subjectivity and inefficiency. However, AI and ML can automate this task by learning to extract intricate features such as melodic scales, rhythm, and ornamentation from music. Through training on extensive datasets of labeled Raga recordings, these algorithms discern unique patterns that distinguish one Raga from another. This not only expedites the analysis but also reveals subtle features that may be challenging for human experts to consistently identify.

AI contributes to reducing subjectivity in Raga detection, ensuring a more objective analysis compared to human interpretation. Collaboration between musicologists, AI researchers, and musicians is crucial for developing models that capture the diverse and culturally nuanced nature of Ragas across various genres and regions. This collaborative effort enhances the accuracy and applicability of these technologies in the dynamic realm of music analysis.

References

[1] Pendyala, Vishnu S. and Yadav, Nupur and Kulkarni, Chetan and Vadlamudi, Lokesh,"Towards Building an Automated Indian Classical Music Tutor for the Masses,", SSRN Electron. J., pp. 200042, 2022.

[2] Singh, Yeshwant, and Anupam Biswas. "Deep learning based Tonic identification in Indian Classical Music," in 5th Conference on Information and Communication Technology (CICT), Kurnool, India, December 10-12, 2021,pp. 1-6

[3] Sinith, M. S. and K. Rajeev. "Pattern Recognition in South Indian Classical Music Using a Hybrid of HMM and DTW." International Conference on Computational Intelligence and Multimedia Applications,(ICCIMA), Sivakasi, Tamil Nadu, December 13-15, 2007, pp. 339-343.

[4] P. Chordia and S. Şentürk, "Joint Recognition of Raag and Tonic in North Indian Music," in Computer Music Journal, vol. 37, no. 3, pp. 82-98, Sept. 2013

[5] D. De and S. Roy, "Inheritance in Indian Classical Music: An object-oriented analysis and pattern recognition approach," 2012 International Conference on Radar, Communication and Computing (ICRCC), Tiruvannamalai, India, 2012, pp. 193-198,

[6] Guo, Gongde, Hui Wang, David A. Bell, Yaxin Bi and Kieran R. C. Greer. "KNN Model-Based Approach in Classification.", OTM Conferences / Workshops,2003

[7] Cunningham, Pádraig and Sarah Jane Delany. "k-Nearest Neighbour Classifiers - A Tutorial", ACM Computing Surveys, Vol. 54,2020,pp. 1-25.

[8] Joshi, Dipti, Jyoti Pareek and Pushkar Ambatkar. "Indian Classical Raga Identification using Machine Learning,", International Semantic Intelligence Conference, New Delhi, India,2021,pp. 259-263

[9] H. Sharma and R. S. Bali, "Comparison of ML classifiers for Raga recognition," Int. J. Sci. Res. Publ., vol. 5, no. 10, 2015, pp. 1–5

[10] A. Anand, "Raga Identification Using Convolutional Neural Network," Second International Conference on Advanced Computational and Communication Paradigms (ICACCP), Gangtok, India, 2019, pp. 1-6, doi: 10.1109/ICACCP.2019.8882942.

[11] Tim Hillel, Michel Bierlaire, Mohammed Z.E.B. Elshafie, Ying Jin,"A systematic review of machine learning classification methodologies for modelling passenger mode choice",Journal of Choice Modelling,Volume 38,2021,100221,ISSN 1755-5345,

[12] Dutta, Shrey, PV KrishnarajSekhar and Hema A. Murthy. "Raga Verification in Carnatic Music Using Longest Common Segment Set." International Society for Music Information Retrieval Conference, Spain,2015, pp. 605-611

[13] R. Taha, S. Alshakrani and N. Hewahi, "Exploring Machine Learning Classifiers for Medical Datasets," 2021 International Conference on Data Analytics for Business and Industry (ICDABI), Sakheer, Bahrain, 2021, pp. 255-259, doi: 10.1109/ICDABI53623.2021.9655862.

[14] Arzaghi, Saman. (2020). Audio Pre-Processing For Deep Learning.

[15] Natsiou, Anastasia and Seán O'Leary. "Audio representations for deep learning in sound synthesis: A review." 2021 IEEE/ACS 18th International Conference on Computer Systems and Applications (AICCSA), Tangier, Morocco, 2021,pp. 1-8.

[16] Choi, Keunwoo, György Fazekas, Kyunghyun Cho and Mark B. Sandler. "A Comparison of Audio Signal Preprocessing Methods for Deep Neural Networks on Music Tagging." ,26th European Signal Processing Conference (EUSIPCO), Greek island of Kos,2017,pp. 1870-1874.

[17] Katti, A., Sumana, M. " Pipeline for Pre-processing of Audio Data", in IOT with Smart Systems. Smart Innovation, Systems and Technologies, vol 312. Springer, Singapore.

[18] Sakshi Indolia, Anil Kumar Goswami, S.P. Mishra, Pooja Asopa,"Conceptual Understanding of Convolutional Neural Network-A Deep Learning Approach",Procedia Computer Science, Volume 132,2018,Pages 679-688,ISSN 1877-0509,

[19] S. Albawi, T. A. Mohammed and S. Al-Zawi, "Understanding of a convolutional neural network," in International Conference on Engineering and Technology (ICET), Antalya, Turkey, 2017, pp. 1-6

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